

UNIVERSITY OF MANITOBA
 Department of Mathematics
Math 1210, Term test 1
May 17, 2023, 10:45 am - 11:35 am
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Grading table

Question:	1	2	3	4	5	Total
Points:	10	8	8	5	7	38
Score:						

1. (10 points) Use mathematical induction to prove that for any integer $n \geq 1$

$$1 + 3 + 5 + 7 + \cdots + (4n + 1) = (2n + 1)^2.$$

Solution: We denote by $P(n)$ the statement

$$1 + 3 + 5 + 7 + \cdots + (4n + 1) = (2n + 1)^2.$$

Let us first verify that $P(1)$ is true. The left-hand side of the equation is the sum $1 + 3 + 5 = 9$. The right-hand side is $(2 \cdot 1 + 1)^2 = 3^2 = 9$. Hence $P(1)$ is true.

Let us now assume that $P(k)$ is true for some integer k , that is

$$1 + 3 + 5 + 7 + \cdots + (4k + 1) = (2k + 1)^2. \quad (1)$$

We want to show that $P(k + 1)$ is also true, that is

$$1 + 3 + 5 + 7 + \cdots + (4k + 5) = (2k + 3)^2. \quad (2)$$

We rewrite the left-hand side of (2) as:

$$\begin{aligned}
 \underbrace{1 + 3 + 5 + 7 + \cdots + (4k + 1)}_{\text{LHS of (1)}} + (4k + 3) + (4k + 5) &= (2k + 1)^2 + (4k + 3) + (4k + 5) \\
 &= (4k^2 + 4k + 1) + 8k + 8 \\
 &= 4k^2 + 12k + 9 \\
 &= (2k + 3)^2.
 \end{aligned}$$

This is the right-hand side of equation (2). Hence we have shown that $P(k + 1)$ is true under the assumption that $P(k)$ is true.

By the principle of mathematical induction, the statement $P(n)$ is true for any $n \geq 1$.

2. (8 points) Use induction to prove that 5 divides $9^{2n} - 6^n$ for any integer $n \geq 1$.

Solution: We denote by $P(n)$ the statement that 5 divides $9^{2n} - 6^n$. When $n = 1$, we see that 5 divides $9^2 - 6 = 81 - 6 = 75$. Hence $P(1)$ is true.

Now suppose that $P(k)$ is true for some integer $k \geq 1$, that is 5 divides $9^{2k} - 6^k$. We want to prove that the statement $P(k+1)$ is true, that is that 5 divides $9^{2k+2} - 6^{k+1}$. We write

$$\begin{aligned} 9^{2k+2} - 6^{k+1} &= 9^{2k} \cdot 9^2 - 6^k \cdot 6 \\ &= (9^{2k} - 6^k + 6^k)9^2 - 6^k \cdot 6 \\ &= (9^{2k} - 6^k)9^2 + 6^k(9^2 - 6). \end{aligned}$$

We know that 5 divides $9^{2k} - 6^k$ by our assumption, and that 5 divides $9^2 - 6 = 75$ by the base case. Hence 5 divides $9^{2k+2} - 6^{k+1}$ and $P(k+1)$ is true.

By the principle of mathematical induction, the proposition $P(n)$ is true for any $n \geq 1$.

3. (a) (5 points) Use the formula $1 + 3 + 5 + 7 + \cdots + (4n + 1) = (2n + 1)^2$ of question 1 to evaluate the sum

$$\sum_{k=2}^{12} (2k + 3).$$

Simplify your result as much as possible.

Solution: We compute:

$$\begin{aligned} \sum_{k=2}^{12} (2k + 3) &= \sum_{k=2}^{12} (2k + 1 + 2) \\ &= \sum_{k=2}^{12} (2k + 1) + \sum_{k=2}^{12} 2 \\ &= \sum_{k=2}^{12} (2k + 1) + 2 \cdot 11 \\ &= \sum_{k=0}^{12} (2k + 1) - 1 - 3 + 2 \cdot 11 \\ &= \sum_{k=0}^{12} (2k + 1) + 18. \end{aligned}$$

By question 1 we know that

$$\sum_{k=0}^{12} (2k + 1) = 1 + 3 + 5 + \cdots + 25 = (2 \cdot 6 + 1)^2 = 13^2 = 169.$$

Thus, it follows that

$$\sum_{k=2}^{12} (2k + 3) = 169 + 18 = 187.$$

- (b) (3 points) Write the sum $-1 + 2 + 5 + 8 + 11 + 14 + \cdots + 41$ in sigma notation starting at the integer 1. Do not evaluate.

Solution: The sum is the sum of integers of the form $3j - 1$ with j going from 0 to 14. Hence

$$\begin{aligned} -1 + 2 + 5 + 8 + 11 + 14 + \cdots + 41 &= \sum_{j=0}^{14} (3j - 1) \\ &= \sum_{k=1}^{15} (3k - 4) \end{aligned}$$

where in the second equality we did the substitution $k = j + 1$.

4. (5 points) Evaluate the expression

$$i^{44} \frac{1 + 3i}{1 - 3i} + 2 \cdot i^{17} + (1 - i)^2.$$

The solution should be in Cartesian form and simplified as much as possible.

Solution: First note that $i^{44} = 1$ and $i^{17} = i$. We compute

$$\begin{aligned} i^{44} \frac{1 + 3i}{1 - 3i} + 2 \cdot i^{17} + (1 - i)^2 &= \frac{1 + 3i}{1 - 3i} + 2i + (1 - i)^2 \\ &= \frac{(1 + 3i)(1 + 3i)}{(1 - 3i)(1 + 3i)} + 2i + (1^2 - 2i + i^2) \\ &= \frac{1^2 + 6i + 9i^2}{1^2 + 3^2} + 2i - 2i \\ &= \frac{-8 + 6i}{10} \\ &= -\frac{2}{5} + \frac{3}{5}i. \end{aligned}$$

5. (7 points) Find all the real numbers t for which the complex number

$$z = 6 \frac{it - 1}{2 - it}$$

has imaginary part equal to 1.

Solution: First we bring z in Cartesian form:

$$\begin{aligned} 6 \frac{it - 1}{2 - it} &= 6 \frac{(it - 1)(2 + it)}{(2 - it)(2 + it)} \\ &= 6 \frac{-t^2 - 2 + it}{4 + t^2} \\ &= -6 \frac{t^2 + 2}{4 + t^2} + i \frac{6t}{4 + t^2}. \end{aligned}$$

Its imaginary part is $6t/(4 + t^2)$ and is equal to 1 exactly when $6t = 4 + t^2$, or equivalently when

$$t^2 - 6t + 4 = 0.$$

This equations has two solutions

$$t = \frac{6 \pm \sqrt{6^2 - 4 \cdot 4}}{2} = \frac{6 \pm \sqrt{20}}{2} = 3 \pm \sqrt{5}.$$

Hence, when $t = 3 \pm \sqrt{5}$, the complex number z has imaginary part 1.